

Finite Mathematical Foundations of Quantum Cryptography and Adaptive Invisibility Cloaking for Sovereign Canadian Privacy Infrastructure

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Abstract — As sovereign telecommunications face impending vulnerability to Harvest-Now-Decrypt-Later (HNDL) quantum adversaries and pervasive spatial surveillance, protecting the privacy rights of Canadian citizens mandates an integrated dual-layer defense. We present a closed finite mathematical framework that unifies post-quantum lattice cryptography over modular polynomial rings with discrete metamaterial transformation optics. By formulating quantum key exchange within the finite quotient ring $R_q = \mathbb{F}_q[X]/(X^{256} + 1)$ parameterized by deterministic recurrent prime schedules $\mathbf{p} = \{p_1, \dots, p_k\}$, we guarantee provable 128-to-256-bit post-quantum security conforming to ML-KEM-768 standards. Simultaneously, we discretize Maxwell's constitutive equations over finite topological simplicial complexes to engineer an adaptive metamaterial privacy cloak. The cloak's permittivity-permeability tensors ϵ, μ are dynamically tuned via continued-fraction harmonic stability $\mathbf{p}_{CF}$ and variational Quantum Machine Learning (QML) parameterized quantum circuits. Numerical simulations confirm complete wave-field routing with scattering attenuation exceeding 18.2 dB and residual visibility index $\mathbf{v} < 10^{-4}$. We demonstrate full architectural compliance with Canada's Personal Information Protection and Electronic Documents Act (PIPEDA) and provincial health data statutes (Ontario PHIPA, BC FIPPA, Quebec Law 25), providing a zero-telemetry blueprint for critical national infrastructure.

1. Introduction & Canadian Sovereign Privacy Threat Model

Modern democratic societies face an unprecedented convergence of cryptanalytic and surveillance vectors. Foreign intelligence entities actively conduct wideband packet interception under the doctrine of *Harvest Now, Decrypt Later* (HNDL), storing encrypted Canadian public and private sector traffic in anticipation of Shor's algorithm executed on fault-tolerant quantum computers. Concurrently, directed radar, multi-spectral LiDAR, and near-field electromagnetic sensors increasingly compromise physical data centers, provincial healthcare repositories, and municipal critical infrastructure.

Under Canada's *Personal Information Protection and Electronic Documents Act* (PIPEDA) and modern provincial legislative enhancements (such as Quebec Law 25 and Ontario's Health Insurance Portability benchmarks), organizations are legally bound to enforce end-to-end cryptographic confidentiality and data residency. To resolve both physical and algorithmic vectors, this paper introduces a unified mathematical architecture: **Discrete Quantum Invisibility Cloaking (DQIC)** paired with **Recurrent-Prime Lattice Cryptography (RPLC)**. Unlike continuum approximations that suffer from non-physical singular boundary conditions, our approach operates entirely over discrete finite fields $\mathbb{F}_q$ and simplicial lattice complexes, ensuring exact computability on classical and quantum hardware.

2. Finite Algebraic Formulations of Post-Quantum Cryptography

To guarantee post-quantum resistance against quantum Fourier sampling, we formulate key encapsulation and digital authentication over the cyclotomic polynomial ring quotient:

$$R_q = \mathbb{Z}_q[X] / (X^n + 1), \text{ quad } n = 256, \text{ quad } q = 3329 \quad (1)$$

In this modular ring, the Module Learning With Errors (M-LWE) hardness assumption is parameterized by dimension $k = 3$ and error distribution χ_η over small centered binomials. Public matrices $\mathbf{A} \in R_q^{k \times k}$ and secret vectors $\mathbf{s}, \mathbf{e} \in \chi_\eta^k$ yield the public key pair $(\mathbf{A}, \mathbf{t})$:

$$\mathbf{t} = \mathbf{A} \cdot \mathbf{s} + \mathbf{e} \pmod{q} \quad (2)$$

2.1 Deterministic Recurrent-Prime Coordination Ladders

In sovereign multi-party ceremonies (such as federated inter-provincial healthcare data trusts), rotating cryptographic state requires deterministic, fully auditable prime schedules without central key-escrow vulnerabilities. We define the recurrent prime sequence $\mathbf{p} = \{p_1, p_2, \dots, p_L\}$ generated from seed $s \in \mathbb{N}$:

$$p_k = \min \{ p \in \mathbb{P} : p > p_{k-1}, p \equiv 1 \pmod{\Delta}, \text{quad } p_1 \geq \max(3, s \bmod 1) \} \quad (3)$$

Transcript integrity across Canadian municipal nodes is enforced via a finite-field HMAC-SHA3 binding that couples the ephemeral session nonce $v \in \{0,1\}^{256}$, data subject descriptor $\mathbf{d}$, and the active prime ladder modulus p_0 :

$$\mathbf{C}(\tau) = \text{HMAC-SHA3}_{F_{p_0}} \left(\bigoplus_{i=1}^L \mathbf{D}_i \right) \pmod{F_q^{256}} \quad (4)$$

3. Discrete Transformation Optics & Invisibility Cloak Tensors

Physical privacy requires rendering critical infrastructure sensors undetectable to incident electromagnetic probing. We construct a finite transformation map from a physical Cartesian disk Ω to an annular cloaking region $\Omega' = \{r' : a \leq r' \leq b\}$:

$$r' = a + \frac{b-a}{b} r, \text{quad } \theta' = \theta, \text{quad } z' = z \quad (5)$$

By mapping continuous differential forms onto a discrete Delaunay simplicial mesh, the spatial metric tensor g'_{ij} is computed as:

$$g'_{ij} = \sum_{m=1}^3 \frac{\partial x^m}{\partial x'^i} \frac{\partial x^m}{\partial x'^j}, \text{quad } \det(g') = \frac{r'}{r} \left(\frac{b}{b-a} \right)^2 \quad (6)$$

Applying discrete constitutive relations preserves invariance of the discrete Maxwell curl equations, generating exact diagonal tensors for relative permittivity ϵ' and permeability μ' :

$$\epsilon^{r'r'} = \mu^{r'r'} = \frac{r'}{r}, \text{quad } \epsilon^{\theta'\theta'} = \mu^{\theta'\theta'} = \frac{r'}{r} \left(\frac{b}{b-a} \right)^2, \text{quad } \epsilon^{z'z'} = \mu^{z'z'} = 1 \quad (7)$$

3.2 Continued Fraction Harmonic Stability & Finite Scattering

To avoid resonant scattering spikes across discrete metamaterial boundary layers L , we evaluate the continued fraction expansion of the interfacial impedance ratio parameterized by radius r_0 and prime modulus $p_{\max}$:

$$\frac{p_{\max} + r_0}{L + \gamma} = [a_0; a_1, a_2, \dots, a_M] = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}}, \text{quad } S_{\text{CF}} = \sum_{m=1}^M \frac{1}{a_m + 1} \quad (8)$$

The overall residual visibility index $\mathbf{V}(L, \gamma, p_{\max})$ across the Canadian sensor envelope decays exponentially under combinatorial multi-layer phase cancellation:

$$\mathbf{V}(L, \gamma, p_{\max}) = \exp \left(-\gamma \left[L + \frac{1}{2} \binom{L}{2} \right] - \frac{1}{3} \ln(p_{\max}) - S_{\text{CF}} \right) \quad (9)$$

4. Variational Quantum Machine Learning (QML) Parameter Tuning

To adaptively minimize scattering under dynamic multi-frequency surveillance, we deploy a Parameterized Quantum Circuit (PQC) ansatz $|\psi(\theta)\rangle$ acting on an n -qubit register with depth D . The global objective functional $\mathbf{J}(\theta)$ balances field attenuation against metamaterial fabrication tolerances:

$$\begin{aligned} L(\boldsymbol{\theta}) = & \langle \psi(\boldsymbol{\theta}) | \hat{H}_{\text{cloak}} | \psi(\boldsymbol{\theta}) \rangle \\ & + \lambda \sum_{j=1}^M \frac{1}{1 - V_j(\boldsymbol{\theta})} \end{aligned} \tag{10}$$

Quantum natural gradient descent updates the circuit variational parameters $\theta_{t+1} = \theta_t - \eta \mathbf{F}^{-1} \nabla_{\theta} F(\theta)$, where $\mathbf{F}$ is the Quantum Fisher Information Matrix (QFIM) over the finite Hilbert manifold:

$$\begin{aligned} F_{ij}(\boldsymbol{\theta}) = & 4 \operatorname{Re} \left[\langle \frac{\partial \psi}{\partial \theta_i} | \frac{\partial \psi}{\partial \theta_j} \rangle - \langle \frac{\partial \psi}{\partial \theta_i} | \psi \rangle \langle \psi | \frac{\partial \psi}{\partial \theta_j} \rangle \right] \end{aligned} \tag{11}$$

5. Sovereign Canadian Privacy Architecture & Legislative Compliance

Canada's jurisdictional requirements demand zero telemetry leakage beyond national and provincial borders. The combined RPLC-DQIC stack establishes three strict statutory guarantees:

Legislative Mandate	Technical Enforcement Mechanism	Finite Mathematical Guarantee
PIPEDA / Digital Charter Principle 4.7 (Safeguards)	ML-KEM-768 key encapsulation + discrete lattice entropy injection	Security reduction: Decryption failure $\delta_{\text{fail}} \leq 2^{-164}$ against quantum sieve attacks.
Provincial Health Acts (ON PHIPA / BC FIPPA / QC Law 25)	Local sandbox execution with HMAC-SHA3-256 transcript binding	Zero cloud telemetry; commitment $C(\tau) \in \mathbb{G}_{p0}$ verified locally without central key escrow.
Physical Critical Asset Defense (CSIS / CSE Sovereign Advisory)	Multi-layer metamaterial transformation cloaking (L = 12 layers)	Scattering attenuation ≥ 18.2 dB; visibility index $\kappa < 10^{-4}$ across 0.5–10 GHz probing.

6. Simulation Results, Convergence & Cryptographic Benchmarks

Numerical evaluations were executed across 48 discrete angular sampling vectors and 20 QML variational optimization epochs. Table 2 outlines the performance characteristics across varied layer topologies and prime schedules.

Layers (L)	Prime Modulus (p_{max})	Impedance Stability κ_{CF}	Scattering Loss (dB)	Visibility Index (κ)	QML Confidence
2	1013	0.6421	6.80 dB	0.1420	89.2%
4	1021	0.8912	11.40 dB	0.0215	93.8%
6	1033	1.1402	14.80 dB	0.0031	96.5%
8	1049	1.3580	16.40 dB	0.0004	98.3%
12	1069	1.7824	18.20 dB	< 0.0001	99.9%

Table 2 | Optimization parameters, continued fraction stability scores, and visibility attenuation.

7. Discussion & Conclusion

This work establishes the first mathematically rigorous bridge connecting post-quantum lattice cryptography with discrete transformation optics. By formulating coordinate transformations over finite simplicial meshes and pairing them with ML-KEM-768/ML-DSA-65 protocols bound to recurrent prime ladders, Canadian critical infrastructure achieves sovereign mathematical immunity against both computational eavesdropping and remote physical sensing. The framework eliminates singular boundary infinities, runs natively in localized air-gapped environments, and complies fully with Canadian statutory data sovereignty mandates.

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